

Gauge Structure and Certified Non-Extendability for Selected Karlsson Hadamards in Dimension Six

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Abstract

Within Karlsson’s three-parameter family $K_6^{(3)}$ only, I prove a gauge-structure theorem, give a certified fourth-MUB obstruction at seven Dita- λ points, independently re-verify one Brierley–Weigert instance by exact Gröbner methods, and correct a literature transcription of Karlsson’s A -matrix. **(T1, Theorem 8):** at $\theta=0$, ϕ is a parametrization gauge; on the Dita slice, λ is not a matrix-level gauge, while CHM class is π -periodic (four classes among $\{0, 0.4, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$). **(T3, Theorem 10; lead result):** at seven Dita- λ points $\lambda \in \{0, 0.4, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$, `HomotopyContinuation certify()` yields a numerical-interval certificate that the single-vector witness W_1 is empty at every recovered-pool 6-clique (40/40), which implies no fourth MUB at those instances. **(T4, Theorem 11; pipeline validation):** at the D_0 -equivalent pair (D_{bc}, F_D) , an exact ideal-membership proof shows that the witness ideal W_1 is the unit ideal over $\mathbb{Q}(\zeta_{24}, \sqrt{5})$ (Gröbner basis $\{1\}$). This independently re-checks a fact already in Brierley–Weigert [6]; it is not billed as a co-equal headline with T1/T3. The closed-form third basis F_D follows Bengtsson–Bruzda–Ericsson–Larsson–Tadej–Życzkowski [5]. **(C1)** The McNulty–Weigert printed A -matrix differs from Karlsson’s by a leading-sign error and a broken conjugate structure; symbolically $A_{MW}A_{MW}^\dagger \neq 2I$ on a Zariski-open set (corroborated by a 0/349 numerical audit). Third-MUB locus identification, search-set audits, and reproducibility artifacts are documented in a companion computational report [1].

Index Terms

Complex Hadamard matrices, Karlsson family, mutually unbiased bases, homotopy continuation, Gröbner bases, certified computation

I. INTRODUCTION

Mutually unbiased bases (MUBs) in $\mathbb{C}^d$ are orthonormal bases $\mathcal{B}_0, \dots, \mathcal{B}_{m-1}$ such that for all $a \neq b$ and all $|\psi\rangle \in \mathcal{B}_a$, $|\phi\rangle \in \mathcal{B}_b$,

$$|\langle\psi|\phi\rangle|^2 = \frac{1}{d}. \quad (1)$$

MUBs play a central role in quantum state tomography, cryptography, and the study of finite geometry [4], [6]. For prime-power dimensions $d = p^n$, exactly $d + 1$ MUBs exist [9], [10]. For composite d , the maximum number $N(d)$ of MUBs is unknown in general; in dimension six, Zauner conjectured $N(6) = 3$ [2], [3].

A standard construction route passes through complex Hadamard matrices (CHMs): unimodular $d \times d$ matrices H with $HH^\dagger = dI$. Karlsson [11] parametrized a three-dimensional family $K_6(\theta, \phi, \lambda)$ of 6×6 CHMs. Grassl [8] showed that $\{I, F_6\}$ extends to at most three MUBs, so searches must be *per* Hadamard matrix. Brierley–Weigert [6] constructed all MU vectors at the Dita matrix D_0 in closed form and showed that none of the ten third bases are pairwise MU.

This paper is theorem-first within $K_6^{(3)}$ only. The lead result is a certified fourth-MUB obstruction at seven Dita- λ points (T3). Supporting algebra is the gauge structure of Karlsson parameters (T1). An exact Gröbner re-verification at one D_0 -equivalent pair (T4) validates the witness pipeline against Brierley–Weigert but does not extend that literature. A literature correction (C1) records that Karlsson’s original A -matrix, not the McNulty–Weigert transcription, produces valid CHMs in numerical tests. The computational identification of third-MUB loci (Dita circle; F_6 $\theta=0$ arc), the covering region $\mathcal{R}$, and reproducibility audits appear in the companion methods report [1]; the loci used to instantiate B_3 in the T3 runs are documented there.

A. Contributions

- T3 Lead:** certified W_1 -emptiness (`HomotopyContinuation certify()`, tolerance-based) at seven Dita- λ points, 40/40 recovered-pool 6-cliques (Theorem 10).
- T1** Gauge structure: ϕ -gauge at $\theta=0$; λ -periodicity; Dita block specialization; Dita λ matrix moduli with π -periodic CHM class (Theorem 8).
- C1** Literature correction: explicit algebraic discrepancy between Karlsson’s A and the McNulty–Weigert transcription (Section V-A).
- T4** Pipeline validation: exact unit-ideal certificate for W_1 at (D_{bc}, F_D) , subordinate to Brierley–Weigert [6] (Theorem 11).

II. SCOPE AND LOGICAL STATUS OF RESULTS

All results concern Karlsson’s family $K_6(\theta, \phi, \lambda)$ only; CHMs outside $K_6^{(3)}$ are not treated.

- **Proved (algebra).** Theorem 8 (T1); symbolic non-unitarity of the printed McNulty–Weigert A -block (C1).
- **Certified (numerical interval).** Theorem 10 (T3): `certify()` over a square subsystem of W_1 at seven listed points; not a certificate on the full λ -circle, on $\mathcal{R}$, or on $N(6)$.
- **Exact ideal membership (one point).** Theorem 11 (T4): Gröbner basis $\{1\}$ over $\mathbb{Q}(\zeta_{24}, \sqrt{5})$ at (D_{bc}, F_D) ; independent re-verification of a BW fact, not a new extendability theorem on the circle.
- **Companion only (reproducibility).** Third-MUB locus identification, $\mathcal{S}^*$ sweeps, grid-miss narrative, pool-completeness tables, and the Open Problem on all of $\mathcal{R}$ [1]. Theorem 10 does not logically depend on that Open Problem.

This paper does not prove Zauner’s conjecture $N(6)=3$ or fourth-MUB absence on all of $K_6^{(3)}$.

MAIN THEOREM

Lead result (Theorem 10). For each

$$\lambda \in \Lambda_{\text{cert}} := \left\{ 0, 0.4, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3} \right\}$$

and every size-6 clique B_3 in the recovered MU pool of $\{\mathbf{I}, H(\theta_D, \pi/4, \lambda)\}$ with $\theta_D = \arccos(1/\sqrt{3})$, the single-vector witness $W_1(H, B_3)$ is empty under HomotopyContinuation interval `certify()` of a generic square subsystem (mixed volume 252), so $\{\mathbf{I}, H, B_3\}$ admits no fourth MUB.

40/40 certified obstructions (native all-pool-clique exhaustion at the seven listed λ).

Certificate details—including the square-subsystem completeness bridge ($N_{\text{expected}} = N_{\text{tracked}} = N_{\text{certified}} = 252$, $N_{\text{full-system}} = 0$)—are in Appendix C; they are not deferred to the companion report.

III. BACKGROUND

A. MUBs From Hadamard Matrices

Let H be a 6×6 CHM with $|H_{ij}| = 1$ and $HH^\dagger = 6I$. The normalized columns $c_k = H_{:,k}/\sqrt{6}$ form an orthonormal basis $\mathcal{B}_H$ satisfying (1) with the computational basis $\mathcal{B}_I = \{e_0, \dots, e_5\}$. A third MUB requires six unit vectors pairwise orthogonal and unbiased to both $\mathcal{B}_I$ and $\mathcal{B}_H$; a fourth MUB requires a second disjoint six-vector set with the same property relative to that third basis.

B. Karlsson’s Three-Parameter Family

Following Karlsson [11] and McNulty–Weigert [13],

$$K_6(\theta, \phi, \lambda) = \begin{bmatrix} F_2 & Z_1 & Z_2 \\ Z_3 & \frac{1}{2}Z_3AZ_1 & \frac{1}{2}Z_3BZ_2 \\ Z_4 & \frac{1}{2}Z_4BZ_1 & \frac{1}{2}Z_4AZ_2 \end{bmatrix}, \quad (2)$$

where $F_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $B = -F_2 - A$, and Z_j are 2×2 blocks from unimodular parameters $z_1, \dots, z_4$ with $z_1 = e^{i\lambda}$. The block A depends on (θ, ϕ) . Karlsson’s original A -matrix formulas are used throughout; the McNulty–Weigert printed transcription is algebraically invalid as a CHM block (Section V-A; contribution C1).

IV. METHODS

A. Per- H MU pool and fourth-MUB witness

For fixed H , seek unit vectors unbiased to $\mathcal{B}_I$ and $\mathcal{B}_H$. With gauge $v_0 = 1$ and $z_i = v_i\sqrt{6}$, unbiasedness to the five independent columns of H together with unimodular auxiliaries $z_i w_i = 1$ yields a square 10×10 polynomial system depending on H . Solutions on the conjugate locus are mapped to unit vectors, verified at tolerance 10^{-8} , and deduplicated modulo global phase; the resulting finite set is the recovered MU pool $\mathcal{P}$. A third basis B_3 is any orthonormal 6-frame whose columns form a size-6 clique in the orthogonality graph on $\mathcal{P}$ (edge when $|\langle v, w \rangle| < \varepsilon_{\text{ortho}}$). The single-vector witness $W_1(H, B_3)$ encodes existence of a unit vector unbiased to $\mathbf{I}$, H , and B_3 (17 equations in 10 variables after unimodular auxiliaries; $n_{\text{wit}} = 1$). Emptiness of W_1 implies no fourth MUB at that (H, B_3) . (Full pipeline engineering, search sets, and reproducibility pins appear in [1]; they are not required to read the T3 certificate.)

TABLE I
CORRECTION SUMMARY FOR THE A -MATRIX TRANSCRIPTION.

	Karlsson 2011	McNulty–Weigert (as printed)
Symbolic A formula	valid	leading $+1/2$; non-conjugate structure
$AA^\dagger = 2I$	identity	fails: $(AA^\dagger - 2I)_{11} = \sqrt{3} \sin \phi \sin \theta$
349-point numerical audit	348/349 pass	0/349 pass
Identified error	—	Eqs. (7.3)–(7.5) as printed

TABLE II
NUMERICAL AUDIT OF A -MATRIX VARIANTS ON 349 PARAMETER POINTS (CORROBORATION OF LEMMA 1). PASS CRITERIA: $\|AA^\dagger - 2I\|_\infty < 10^{-8}$, $\|BB^\dagger - 2I\|_\infty < 10^{-8}$, VALID CHM, MÖBIUS CONSISTENCY $< 10^{-6}$.

Variant	Pass	Fail	Notes
Karlsson 2011 (original)	348	1	One NaN at degenerate Möbius point
McNulty–Weigert (as printed)	0	349	A not unitary; CHM invalid

B. T3 certificate (numerical interval)

At each listed λ , every recovered-pool 6-clique is submitted to a generic square subsystem of W_1 (mixed volume 252), HomotopyContinuation path tracking, interval-arithmetic `certify()`, and residual check against the full system (tolerance 10^{-8}). Emptiness means $N_{\text{expected}} = N_{\text{tracked}} = N_{\text{certified}} = 252$ and $N_{\text{full-system}} = 0$ (Appendix C). This is *not* a Gröbner unit-ideal certificate.

C. T4 certificate (exact ideal membership)

At (D_{bc}, F_D) with algebraic entries in $\mathbb{Q}(\zeta_{24}, \sqrt{5})$, Macaulay2 computes a Gröbner basis of the W_1 ideal equal to $\{1\}$ (Section C).

D. Central open obstacle

A theorem of fourth-MUB absence on all of a covering region $\mathcal{R} \subset K_6^{(3)}$ (defined in [1]) would require exact-coefficient elimination with B_3 in a genuine number field at points not CHM-equivalent to D_0 . The reduced two-vector witness run over $\mathbb{CC}$ is inconclusive (inexact-field warning; Remark 2). T4 works precisely because $B_3 = F_D$ is closed-form at a D_0 -equivalent point—corollary-grade relative to Brierley–Weigert as a mathematical result, valuable here as a machine-checkable cross-check. Theorem 10 does not depend on resolving that open obstacle.

V. RESULTS

A. Correction to the McNulty–Weigert A -matrix transcription

Karlsson [11] uses

$$A_{11} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}(\cos \theta + e^{-i\phi} \sin \theta), \quad A_{12} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}(-\cos \theta + e^{i\phi} \sin \theta),$$

with Hermitian block structure $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{12}^* & -A_{11}^* \end{pmatrix}$. Eqs. (7.3)–(7.5) of [13], as printed, instead take leading $+1/2$ in A_{11} and assemble $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{12} & -A_{11} \end{pmatrix}$ (no conjugate on the off-diagonal).

Lemma 1 (Printed McNulty–Weigert A is not unitary). *Let $A_{\text{MW}}(\theta, \phi)$ be the printed block above. Then*

$$(A_{\text{MW}}A_{\text{MW}}^\dagger - 2I)_{11} = \sqrt{3} \sin \phi \sin \theta,$$

which is not identically zero. Hence $A_{\text{MW}}A_{\text{MW}}^\dagger \neq 2I$ on a Zariski-open subset of (θ, ϕ) , so the printed transcription cannot define a valid Karlsson CHM on that open set. Moreover $A_{11}^{(\text{Karlsson})} - A_{11}^{(\text{MW})} = -1$ identically (leading-sign discrepancy).

Proof. Direct expansion of $A_{\text{MW}}A_{\text{MW}}^\dagger$ (computer algebra; reproducible from `scripts/python/audit_karlsson_variants.py` formulas) yields the displayed $(1,1)$ entry. The leading-sign identity is immediate from $\pm\frac{1}{2}$. $\square$

B. Gauge Structure (T1)

Theorem 8 records that at $\theta=0$, ϕ does not enter H ; the family is 2π -periodic in λ ; on the Dita slice λ changes the matrix (Lemma 6) while CHM class is π -periodic with four classes among the seven sample points of Lemma 7. Proof: Appendix A.

C. Certified Fourth-MUB Obstruction at Seven Dita Points (T3)

Theorem 10 is the lead result (Section II): at $\Lambda_{\text{cert}} = \{0, 0.4, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$, `certify()` finds W_1 empty for all recovered-pool 6-cliques (40/40), covering four CHM classes (Lemma 7). Full proof: Appendix A. Confirmatory dense probes and search-set negatives outside these seven points are reproducibility material in [1]; they are not part of T3.

D. Pipeline Validation: Exact Re-Verification at the D_0 -Equivalent Pair (T4)

Theorem 11 establishes $W_1(D_{\text{bc}}, F_D) = \langle 1 \rangle$ by Gröbner basis over $\mathbb{Q}(\zeta_{24}, \sqrt{5})$. Brierley–Weigert already proved the stronger complete-pool statement ($N_v=120, N_t=10, N_p=0$); T4 covers one third basis by a different exact method and is billed here as independent pipeline validation (Remark 3).

VI. DISCUSSION

T3 is a pointwise certified obstruction, not a theorem on $\mathcal{R}$ or $N(6)$. T4 does not extend Brierley–Weigert; it cross-checks one algebraic instance of the same witness ideal used numerically in T3. C1 is an algebraic transcription correction (Lemma 1) and should be reported to the authors of [13]. The companion report [1] documents search-set reproducibility and why dense-grid negatives alone cannot promote sweep results to theorems—reinforcing that T3 is the exception that does not rest on grid density.

VII. CONCLUSION

Scoped to $K_6^{(3)}$ only, this paper delivers a gauge theorem (T1), a lead certified fourth-MUB obstruction at seven Dita- λ points (T3), an algebraic literature correction (C1), and an exact pipeline-validation re-check at one D_0 -equivalent pair (T4), subordinate to Brierley–Weigert. I do not claim fourth-MUB absence on all of $K_6^{(3)}$ or on $N(6)$. Closing the gap to a theorem on $\mathcal{R}$ requires algebraic B_3 at points not CHM-equivalent to D_0 ; see [1] for the reproducibility record of that open obstacle.

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APPENDIX A

PROOFS OF T1, T3, AND T4

Theorem and Proof are used for T1 (Lemmas 3–7), T3, and T4.

APPENDIX B

GAUGE STRUCTURE IN KARLSSON’S FAMILY

Write Karlsson’s three-parameter family as $K_6(\theta, \phi, \lambda)$ with block $A(\theta, \phi)$ and $z_1 = e^{i\lambda}$ determining the remaining Möbius blocks (as in Section III). Throughout, $H(\theta, \phi, \lambda) := K_6(\theta, \phi, \lambda)$.

Lemma 2 (Block A at $\theta = 0$). *For all $\phi, \phi' \in [0, \pi)$ and $\lambda \in \mathbb{R}$,*

$$A(0, \phi) = A(0, \phi') = \begin{bmatrix} -\frac{1}{2} + i\frac{\sqrt{3}}{2} & -\frac{1}{2} - i\frac{\sqrt{3}}{2} \\ -\frac{1}{2} - i\frac{\sqrt{3}}{2} & \frac{1}{2} - i\frac{\sqrt{3}}{2} \end{bmatrix}.$$

Proof. Substitute $\sin \theta = 0$ in the Karlsson formulas

$$A_{11} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}(\cos \theta + e^{-i\phi} \sin \theta), \quad A_{12} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}(-\cos \theta + e^{i\phi} \sin \theta).$$

All ϕ -dependent terms vanish, giving the displayed matrix. Unitarity $AA^\dagger = 2I$ follows by direct multiplication. $\square$

Lemma 3 (ϕ -gauge for H at $\theta = 0$). *For all ϕ, ϕ', λ : $H(0, \phi, \lambda) = H(0, \phi', \lambda)$ entrywise.*

Proof. At $\theta = 0$, Lemma 2 gives A and $B = -F_2 - A$ independent of ϕ . The Möbius parameters $\alpha_A, \beta_A, \alpha_B, \beta_B$ depend only on A, B , hence on θ alone. With $z_1 = e^{i\lambda}$, the quantities z_2, z_3, z_4 and all blocks Z_j are ϕ -independent. Therefore the assembled 6×6 matrix $H(0, \phi, \lambda)$ is independent of ϕ . $\square$

Lemma 4 (λ -periodicity). *For all θ, ϕ, λ : $H(\theta, \phi, \lambda + 2\pi) = H(\theta, \phi, \lambda)$.*

Proof. The only λ -dependence enters through $z_1 = e^{i\lambda}$, which satisfies $e^{i(\lambda+2\pi)} = e^{i\lambda}$. All subsequent Möbius/square-root steps depend on z_1^2 and hence are 2π -periodic in λ . $\square$

Lemma 5 (Dita block specialization). *At $\theta = \arccos(1/\sqrt{3})$ and $\phi = \pi/4$,*

$$A(\theta_D, \pi/4) = \begin{bmatrix} i & -1 \\ -1 & i \end{bmatrix}, \quad AA^\dagger = 2I.$$

Proof. With $\cos \theta = 1/\sqrt{3}$, $\sin \theta = \sqrt{2/3}$, and $\phi = \pi/4$,

$$A_{11} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}\left(\frac{1}{\sqrt{3}} - i\frac{\sqrt{2}}{\sqrt{3}}\right) = i, \quad A_{12} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}\left(-\frac{1}{\sqrt{3}} - i\frac{\sqrt{2}}{\sqrt{3}}\right) = -1.$$

Hermitian structure $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{11}^* \end{pmatrix}$ yields the matrix shown. Direct computation gives $AA^\dagger = 2I$. $\square$

Lemma 6 (Matrix inequality in λ at Dita). *Let $\theta_D = \arccos(1/\sqrt{3})$, $\phi = \pi/4$, and write $H(\lambda) := H(\theta_D, \phi, \lambda)$. If $\lambda \not\equiv \lambda' \pmod{2\pi}$, then $H(\lambda) \neq H(\lambda')$ as matrices.*

Proof. At $(\theta_D, \pi/4)$ the block A is fixed (Lemma 5). The only λ -dependence enters through $z_1 = e^{i\lambda}$; distinct z_1 modulo 2π change the Möbius-derived blocks Z_j , so the assembled 6×6 matrices differ entrywise. $\square$

Lemma 7 (CHM class structure on the Dita λ -circle). *Let $\theta_D = \arccos(1/\sqrt{3})$, $\phi = \pi/4$, and $H(\lambda) := H(\theta_D, \phi, \lambda)$.*

- 1) π -periodicity of CHM class. *Dephased alignment over 720 column permutations and diagonal phases yields residuals $\sim 10^{-15}$ for the pairs $(\lambda, \lambda + \pi)$ with $\lambda \in \{0, \pi/3, 2\pi/3\}$ (`chm_equivalence_residual`). Thus $H(\lambda)$ and $H(\lambda + \pi)$ are CHM-equivalent on this slice.*
- 2) Four distinct classes at standard sample points. *The representatives $\{H(0), H(0.4), H(\pi/3), H(2\pi/3)\}$ are pairwise CHM-inequivalent (minimum residual $\gtrsim 0.49$; `results/verify_nwitl_referee_checks.txt`). Combined with item (1), the seven values $\{0, 0.4, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$ cover exactly these four CHM classes.*
- 3) Not a parametrization gauge. *Distinct λ modulo 2π never yield identical matrices (Lemma 6); λ is a genuine moduli coordinate at the matrix level, even though CHM class is π -periodic.*

*In particular, it is **not** claimed that every pair $\lambda \not\equiv \lambda' \pmod{2\pi}$ gives CHM-inequivalent matrices—that would contradict item (1). A complete invariant-theoretic classification of all $\lambda \in [0, 2\pi)$ modulo diagonal-phase/permutation CHM moves remains open; items (1)–(2) record the class structure used by the seven-point certificate below.*

Proof. Item (3) is Lemma 6. Items (1)–(2) are the CHM-equivalence computations recorded in the cited verification logs. $\square$

Theorem 8 (Gauge structure, T1). *Within Karlsson's family $K_6^{(3)}$:*

- 1) *At $\theta = 0$, the coordinate ϕ is a parametrization gauge: $H(0, \phi, \lambda)$ is constant in ϕ (Lemma 3).*
- 2) *The family is 2π -periodic in λ (Lemma 4).*
- 3) *At the Dita slice $(\theta_D, \pi/4)$, the block A is the fixed unitary in Lemma 5.*
- 4) *On the Dita slice, λ is not a matrix-level gauge (Lemma 6), while CHM class is π -periodic with four distinct classes among $\{0, 0.4, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$ (Lemma 7).*

Proof. Immediate from Lemmas 3–7. $\square$

APPENDIX C

FOURTH-MUB OBSTRUCTION ON THE DITA SLICE (T3)

Lemma 9 (CHM equivalence transfers W_1 -emptiness). *Suppose $H_1, H_2 \in K_6^{(3)}$ are CHM-equivalent via diagonal unitaries D_r, D_c and a column permutation P , in the sense that after the standard dephasing gauge*

$$\text{dephase}(H_1) = D_r \text{dephase}(H_2) P D_c.$$

Let B_3 be an orthonormal 6×6 matrix whose columns are mutually unbiased to $\mathbf{I}$ and to the columns of H_2 , and set $B'_3 := D_r B_3$ (columnwise, then renormalized). Then:

- 1) *the columns of B'_3 are orthonormal and mutually unbiased to $\mathbf{I}$ and to the columns of H_1 (hence B'_3 is a third MUB for $\{\mathbf{I}, H_1\}$);*
- 2) *a unit vector v solves $W_1(H_2, B_3)$ if and only if $D_r v$ solves $W_1(H_1, B'_3)$.*

In particular, $W_1(H_2, B_3)$ is empty if and only if $W_1(H_1, B'_3)$ is empty.

Sketch. Diagonal unitaries act entrywise by phases, so $|(D_r v)_i| = |v_i|$ and $\langle D_r u, D_r w \rangle = \langle u, w \rangle$. Mutual unbiasedness to $\mathbf{I}$ and pairwise orthogonality are therefore preserved by $v \mapsto D_r v$ and $B_3 \mapsto B'_3$. For mutual unbiasedness to the Hadamard: the CHM relation identifies columns of (dephased) H_1 with phased, permuted columns of (dephased) H_2 left-multiplied by D_r , so

$$|\langle D_r v, (H_1)_{:,k} \rangle| = |\langle v, (H_2)_{:,p(k)} \rangle|$$

up to the dephasing gauge (itself diagonal phases). Thus MU-to- H and MU-to- B_3 pass to MU-to- H_1 and MU-to- B'_3 , and conversely. $\square$

Theorem 10 (No fourth MUB at certified Dita- λ points). *Let $\theta_D = \arccos(1/\sqrt{3})$, $\phi = \pi/4$, and*

$$\Lambda_{\text{cert}} := \left\{0, 0.4, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}\right\}.$$

Write $H(\lambda) = H(\theta_D, \phi, \lambda)$. For every $\lambda \in \Lambda_{\text{cert}}$ and every third MUB B_3 arising as a size-6 clique in the recovered MU pool of $\{\mathbf{I}, H(\lambda)\}$ (§IV), the triple $\{\mathbf{I}, H(\lambda), B_3\}$ does not extend to a fourth mutually unbiased basis.

Proof. Logic. A fourth MUB $B_4 = \{v_1, \dots, v_6\}$ extending $\{\mathbf{I}, H, B_3\}$ consists of six orthonormal vectors, each mutually unbiased to every column of $\mathbf{I}$, H , and B_3 . After a global phase fixing the first coordinate positive, each v_j is a solution of the single-vector witness $W_1(H, B_3)$: five unimodular gauge equations $|z_i| = 1$ ($i = 1, \dots, 5$) together with mutual-unbiasedness to the six columns of H and of B_3 (17 equations in 10 complex variables; `build_numeric_fourth_mub_witness_system`, $n_{\text{wit}} = 1$). Hence a fourth MUB implies feasibility of W_1 ; contrapositively, emptiness of W_1 proves absence of a fourth MUB at that (H, B_3) .¹

Method (every certified instance). Form a generic square subsystem of W_1 by ten random complex linear combinations of the seventeen equations (Bertini / HomotopyContinuation pattern; fixed seed), compute its mixed volume (252 in every run below), track all 252 start paths, call HomotopyContinuation’s interval-arithmetic `certify()` on every nonsingular endpoint, and residual-check each certified square-root against the full overdetermined system (tolerance 10^{-8}). For each submitted (H, B_3) the accounting is

$$N_{\text{expected}} = N_{\text{tracked}} = N_{\text{certified}} = 252, \quad N_{\text{full-system}} := \text{full_residual_pass} = 0.$$

Square-subsystem completeness bridge. The full witness W_1 is overdetermined (17 equations in 10 complex variables after unimodular auxiliaries). A generic choice of ten complex linear combinations yields a square system whose isolated roots, by the standard Bertini / generic-linear-combination principle used in numerical algebraic geometry, include every isolated solution of the full system that remains isolated under that generic combination. Path tracking recovers all $N_{\text{expected}} = 252$ start paths of the square system (mixed volume 252 throughout); interval-arithmetic `certify()` validates nonsingular endpoints ($N_{\text{certified}} = 252$); residual filtering against the full 17-equation system returns $N_{\text{full-system}} = 0$. Hence no approximate root of the full witness appears among the certified square-system endpoints. This is a numerical-interval certificate of emptiness for that (H, B_3) , *not* a Gröbner unit-ideal membership proof (contrast Theorem 11).

Native all-pool-clique exhaustion at seven λ . At each $\lambda \in \Lambda_{\text{cert}}$, every distinct size-6 clique in the recovered MU pool was submitted to the method above:

λ	$\#\{6\text{-cliques}\}$	empty
0	10	10/10
0.4	4	4/4
$\pi/3$	4	4/4
$2\pi/3$	4	4/4
π	10	10/10
$4\pi/3$	4	4/4
$5\pi/3$	4	4/4

Total: 40/40 empty (results/certify_nwit1_all_cliques_four_classes.txt, results/certify_nwit1_all_cliques_lambdapi.txt). In each run `found_fourth=false` as well.

CHM-class structure (independence, not inheritance). The 720-permutation / diagonal-phase search (`chm_equivalence_residual`) shows $H(\lambda) \sim H(\lambda + \pi)$ on this slice (residuals $\sim 10^{-15}$), while $\{H(0), H(0.4), H(\pi/3), H(2\pi/3)\}$ are pairwise CHM-inequivalent (minimum residual $\gtrsim 0.49$; results/verify_nwit1_referee_checks.txt), as recorded in Lemma 7. The seven points therefore cover four CHM classes; the native 40/40 check is independent at each λ (not deduced from representatives). As corroboration, CHM transport of every clique at $\{0, \pi/3, 2\pi/3\}$ to $\{\pi, 4\pi/3, 5\pi/3\}$ also yields 18/18 empty (results/verify_chm_b3_transfer.txt; Lemma 9).

Clique-index nondeterminism. Among equal-length 6-cliques, Graphs.jl enumeration order need not repeat across re-solves, so a single `argmax(|C|)` index set is not a stable label. The native 40/40 exhaustion removes dependence on that choice by ranging over every size-6 clique in the recovered pool at each λ . $\square$

Remark 1 (Scope — what is and is not claimed). *Theorem 10 is a certified non-existence result at seven explicit Dita- λ values (four CHM classes), with native all-pool-clique coverage at every listed λ (40/40). “Every size-6 clique” means every such clique in the MU pool recovered by the per- H homotopy at that run (pool sizes 120 at $\lambda=0, \pi$ and 72 elsewhere on this list); it does not by itself certify against hypothetical third MUBs missed by an incomplete root track. It is **not** a symbolic identity for all $\lambda \in [0, 2\pi)$, not a theorem on the covering region $\mathcal{R}$ defined in the companion methods report (Paper 2), and not a statement about $N(6)$ or CHMs outside $K_6^{(3)}$. A separate dense probe finds no fourth MUB at 628/628 samples on $[0, 2\pi]$; that numerical evidence is documented in Paper 2 and is **not** part of the certificate in Theorem 10.*

Remark 2 (Pointer to the central open obstacle). *The two-vector Macaulay2 run over CC ($\dim I = -1$ with an inexact-field warning) is **not** a certificate and is **not** confirmatory of T3. T3 uses $n_{\text{wit}} = 1$ (mixed volume 252) instead. The failed*

¹The converse fails in general: W_1 may be feasible without six mutually orthogonal solutions. Emptiness is therefore a complete—indeed strictly stronger—certificate of fourth-MUB non-existence at the point.

exact-elimination attempt is the central obstacle to a theorem on all of $\mathcal{R}$; see §IV-D and the companion methods audit (Paper 2).

Theorem 11 (Exact-coefficient W_1 emptiness at the D_0 -equivalent pair (T4)). *Let D_{bc} be the block-circulant representative of the Dita matrix D_0 and F_D the closed-form third MUB for $\{\mathbf{I}, D_{bc}\}$ (Bengtsson–Bruzda–Ericsson–Larsson–Tadej–Życzkowski 2007, eqs. (79)–(80) and (78),(61)), both with entries in $\mathbb{Q}(\zeta_{24}, \sqrt{5})$. Then the single-vector witness ideal of $W_1(D_{bc}, F_D)$ — seventeen generators (five unimodular relations $z_i w_i = 1$ and twelve mutual-unbiasedness products) in $\mathbb{Q}(\zeta_{24}, \sqrt{5})[z_1, \dots, z_5, w_1, \dots, w_5]$; `symbolic_export/w1_D0_exact.m2` — is the unit ideal. Consequently no flat vector is mutually unbiased to $\mathbf{I}$, D_{bc} , and F_D simultaneously: the MUB triple $\{\mathbf{I}, D_{bc}, F_D\}$ does not extend by even a single vector of a fourth MUB.*

Proof. Logical status. The implication is the same as in Theorem 10 (W_1 empty $\Rightarrow$ no fourth MUB at that (H, B_3)), but emptiness here is established as ideal-theoretic unit-ideal membership over an exact number field — a Gröbner certificate, not a homotopy certify() run.

Field validity. $[\mathbb{Q}(\zeta_{24}, \sqrt{5}) : \mathbb{Q}] = 16$: $\Phi_{24}(x) = x^8 - x^4 + 1$ is irreducible over $\mathbb{Q}$, and 5 is not a square in $\mathbb{Q}(\zeta_{24})$ — certified by the split prime $p = 73$ ($p \equiv 1 \pmod{24}$, so Φ_{24} splits completely mod 73 and $\mathbb{Q}(\zeta_{24})$ embeds in residue fields $\mathbb{F}_{73}$, where 5 is a quadratic non-residue). Hence Macaulay2's toField quotient $\mathbb{Q}[z, s_5]/(\Phi_{24}(z), s_5^2 - 5)$ is genuinely a field and Gröbner arithmetic over it involves no zero divisors (scripts/julia/verify_field_degree_w1.jl; results/track_c_elimination/w1_D0_field_degree.txt).

Method and verification. Macaulay2 1.24.05, bounded cost probe first, then the certificate run:

run	field	result	wall
probe ($\zeta_{24} \mapsto 2, \sqrt{5} \mapsto 103$)	$\mathbb{Z}/241$	unit ideal (signal only)	0.12 s
bounded dim	$\mathbb{Q}(\zeta_{24}, \sqrt{5})$	dim = -1	29.0 s
certificate	$\mathbb{Q}(\zeta_{24}, \sqrt{5})$	GB = {1}, 1 % I = 0	16.0 s

The certificate run computes the full Gröbner basis (gens gb), which is the single element 1, and confirms 1 % I = 0 (symbolic_export/w1_D0_groebner.m2; results/track_c_elimination/w1_D0_groebner.log, w1_D0_cost_estimate.log). No inexact-field warning arises (contrast Remark 2). $\square$

Remark 3 (Scope of Theorem 11 — what is and is not claimed). (i) Point scope. *The exact statement is at the pair (D_{bc}, F_D) itself. Its placement on the Karlsson Dita circle at $\lambda \in \{\pi/2, 3\pi/2\}$ (and nowhere else) rests on a numerical CHM identification of $H(\theta_D, \pi/4, \lambda)$ with D_0/D_{bc} at those two λ (dephased residual $\sim 10^{-13}$; results/identify_d0_in_karlsson.txt). Neither point lies in Λ_{cert} , so Theorems 10 and 11 cover disjoint point sets.*

(ii) Clique scope. *Coverage is the single third basis F_D ; this is **not** the all-pool-clique exhaustion of Theorem 10, and it is not a theorem on the Dita circle, on $\mathcal{R}$, or on all of $K_6^{(3)}$.*

(iii) Object / method / result vs Brierley–Weigert (2009). *Headline: Theorem 11 reproves a fact BW already established, by an independent exact method; it does **not** extend BW. Asymmetry: T4 is weaker in coverage (one third basis F_D vs BW's full pool $N_v=120, N_t=10$) and stronger in one narrow sense (exact symbolic unit-ideal certificate for the fixed- (H, B_3) witness W_1 over $\mathbb{Q}(\zeta_{24}, \sqrt{5})$, versus BW's Groebner-on-pool construction plus combinatorial $N_p=0$). D_{bc} is CHM-equivalent to textbook D_0 , and F_D is Bengtsson's closed-form third MUB—neither object is new. BW's $N_p=0$ already implies $W_1(D_0, B_3) = \emptyset$ for every third basis from the pool, so the mathematical result at this point is corollary-grade; the value here is a machine-checkable cross-check on that instance via a different ideal—a pipeline-validation result, not a co-equal headline with Theorems 8 and 10. The same certificate at $\lambda = 0$ or $\pi/3$ (points not CHM-equivalent to D_0) remains open, blocked on an algebraic form of B_3 there.*

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